



BIOLOGY HSSC-II
SECTION – A (Marks 17)

Time allowed: 25 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

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Deleting/overwriting is not allowed. Do not use lead pencil..

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Cortisol is also called:	Epinephrine	Glucocorticoid	Mineralocorticoid	Androgen	☐	☐	☐	☐
2. Which of the following is in correct order?	i. Trachea ii. Bronchi iii. Alveoli	i. Bronchi ii. pharynx iii. Trachea	i. Alveoli ii. Trachea iii. Bronchi	i. Trachea ii. Pharynx iii. Alveoli	☐	☐	☐	☐
3. Shivering in humans helps to:	Decrease heat production	Decrease body temperature	Increase heat production	Reduce metabolism	☐	☐	☐	☐
4. During muscle contraction, which part of Sarcomere remains constant in length?	Z line	A band	I band	H zone	☐	☐	☐	☐
5. The corpus collosum:	Provides balance and equilibrium in voluntary movements	Is a relay center between Hind and forebrain	Connects the right and the left Hemisphere	Controls auditory and visual reflexes	☐	☐	☐	☐
6. HCl and Intrinsic factors are secreted by the stomach cells:	Chief cells	Parietal cells	Goblet cells	Zymogen cells	☐	☐	☐	☐
7. Which one of the following is the Therapeutic use of monoclonal antibodies?	Targeted cancer therapy	Allergic reactions	Diagnosis of diseases	Blood transfusion reactions	☐	☐	☐	☐
8. The valve that prevents backflow of blood from left ventricle into left atrium:	Aortic valve	Tricuspid valve	Pulmonary valve	Bicuspid valve	☐	☐	☐	☐
9. Tubular secretion involves the movement of substances from:	Ureter to urethra	Filtrate to blood	Blood to filtrate	Kidney to bladder	☐	☐	☐	☐
10. Antibodies transferred from mother to baby through placenta provide:	Natural passive immunity	Artificial active immunity	Natural active immunity	Artificial passive immunity	☐	☐	☐	☐
11. The first and the most important step in polymerase chain reaction:	Blotting	Primer annealing	Primer extension	Denaturation	☐	☐	☐	☐
12. Which organisms are most affected by ocean acidification?	Marine Mammals	Fish	Birds	Shell-forming organisms	☐	☐	☐	☐
13. A major goal of synthetic biology is to:	Prevent mutations in Genome	Eliminate all microorganisms	Create useful biological products and systems	Study ancient DNA only	☐	☐	☐	☐
14. Which one of the following does not correspond to the thin myofilament?	Tropomyosin	Actin	Myosine	Troponin	☐	☐	☐	☐
15. Plasmids are used as vectors because they:	Cannot carry foreign genes	Destroy DNA of virus	Have an origin of replication	Only exist in viruses	☐	☐	☐	☐
16. Range of the data 10, 12, 15, 11, 13, is:	15	2.5	5	10	☐	☐	☐	☐
17. Which of the following is an example of homologous structures?	Fins of fish and wings of birds	Human arm and whale flipper	Wings of birds and insects	Eye of human and eye of octopus	☐	☐	☐	☐



BIOLOGY HSSC-II

38

Time allowed: 2:35 Hours

Total Marks Sections B and C: 68

SECTION – B (Marks 42)

Q. 2 Answer the following parts briefly.

14x3=42

(i)	What is meant by Pyelonephritis?	03	OR	Enlist any three properties of an efficient respiratory surface.	1x3
(ii)	Enlist any three problems associated with kidney transplant.	1x3	OR	Outline the role of respiratory pigments. (Three points)	03
(iii)	Which component of bile helps in digestion? And how? Explain briefly.	1+2	OR	What changes occur in sarcomere during contraction?	03
(iv)	Calculate Median and Mean of given data: 20, 14, 6, 8, 9, 12, 16, 4.	1+2	OR	a. How can ECG help in detecting heart abnormalities? b. State any two conditions that can be diagnosed using ECG.	1+2
(v)	Identify the joint shown in diagram. Also label its parts A, B, C and D.	1+2	OR	Identify and briefly describe the processes X, Y, and Z.	1x3
(vi)	What are macrophages? List any two functions of macrophages in defense.	1+2	OR	a. Describe the process of release of neurotransmitters at the synaptic knob. b. How do neurotransmitters affect the post-synaptic membrane?	2+1
(vii)	Compare arthritis and osteoporosis.	03	OR	a. What is meant by Double-Walled sac in context of pericardium? b. State any two functions of pericardium.	1+2
(viii)	Briefly describe the following: a. Antivirals b. Antiretrovirals	1.5 + 1.5	OR	Why does stomach produce HCl and mucin?	1.5 + 1.5
(ix)	Compare Estrogen and Testosterone.	1.5 + 1.5	OR	State three challenges faced by marine osmoregulators.	1x3
(x)	State any three harmful effects of climate change on ocean biology.	1x3	OR	Briefly explain any three types of T-Cells.	1x3
(xi)	What is meant by kidney failure? Enlist any four causes of chronic renal failure.	1+2	OR	a. What is meant by Genetically Modified Organism? b. Briefly state any two applications of Transgenic bacteria.	1+2
(xii)	Briefly explain any two applications of synthetic biology.	1.5 + 1.5	OR	Write names and briefly discuss any two species that have gone extinct.	1.5 + 1.5
(xiii)	Briefly explain the following and give one example of each: a. Sequence Homology b. Structural Homology	1.5 + 1.5	OR	Give a brief comparison of MRI and CT scan.	1.5 + 1.5
(xiv)	Briefly explain the role of 'receptors' and 'control center' in homeostatic mechanism.	1.5 + 1.5	OR	Briefly discuss the clinical development in process of drug development.	03

SECTION – C (Marks 26)

Note: Attempt the following questions.

Q.3	Explain the structure of nephron. Make diagram as well.	4+3	OR	How sliding and filament model explain muscle control action? Support your answer with suitable diagram.	5+2
Q.4	a. Draw labeled structural model of antibody molecule. b. Differentiate between active and passive immunity.	3+3	OR	Describe three active hormones produced by thyroid gland.	2x3
Q.5	Discuss the path of blood through pulmonary and systemic circulation. Draw diagram as well.	4+3	OR	Explain the sequence of events that take place during swallowing and peristalsis of food.	07
Q.6	Describe the steps involved in Polymerase Chain Reaction PCR. Make flow sheet as well.	5+1	OR	Explain physical responses (Vasodilation and Sweating) of human body to hot conditions. Also make a simple sketch.	4+2



BIOLOGY HSSC-II **SECTION – A (Marks 17)**

Time allowed: 25 Minutes

Candidate Sign. _____

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Answer Sheet No. _____

Signature should not cross this line

Version No.				
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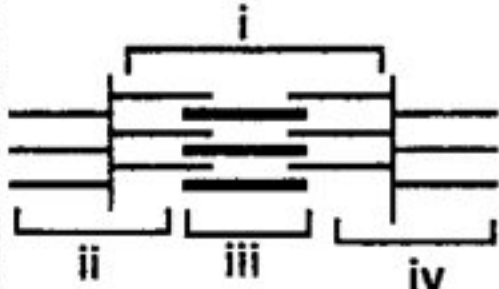
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Fill the relevant bubble against each question according to curriculum:

Question	سوال	A	B	C	D	A	B	C	D
1. Structural homology refers to:		Completely different organs	Unrelated species features	Similar structure due to common ancestry	Similar function but different structure	☐	☐	☐	☐
2. Which one of the following is NOT an endocrine gland?		Pancreas	Parathyroid	Salivary gland	Testis	☐	☐	☐	☐
3. Which structure prevents food from entering the trachea?		Pharynx	Bronchi	Larynx	Epiglottis	☐	☐	☐	☐
4. Which process helps to conserve heat in cold conditions?		Vasoconstriction	Evaporation	Sweating	Vasodilation	☐	☐	☐	☐
5. In the sliding filament model, muscle contraction occurs due to:		Breaking of filaments	Sliding of actin over myosin	Shortening of myosin filaments	Fusion of filaments	☐	☐	☐	☐
6. The sodium-potassium pump actively transports:		3 Na ⁺ out and 2 K ⁺ in	Equal Na ⁺ , K ⁺ in and out	3 Na ⁺ in and 2 K ⁺ out	2 Na ⁺ in and 3 K ⁺ out	☐	☐	☐	☐
7. Which enzyme is primarily responsible for protein digestion in the small intestine?		Lipase	Maltase	Amylase	Trypsin	☐	☐	☐	☐
8. Main advantage of using monoclonal antibodies is:		Target multiple antigens	Low cost of production	Greater specificity and consistency	Longer shelf life	☐	☐	☐	☐
9. Blood enters the right atrium from the body through the:		Vena cava	Pulmonary artery	Pulmonary vein	Aorta	☐	☐	☐	☐
10. Glomerular filtration occurs in the:		Collecting duct	Ureter	Loop of Henle	Bowman's capsule	☐	☐	☐	☐
11. Active immunity is developed by:		Transfer of plasma	Blood transfusion	Exposure to antigens	Injection of antibodies	☐	☐	☐	☐
12. Polymerase chain reaction is a _____ technique.		DNA degradation	DNA hybridization	DNA sequencing	DNA amplification	☐	☐	☐	☐
13. An increase in ocean temperature mainly leads to:		Coral bleaching	Reduced sunlight	Increased salinity	Decreased evaporation	☐	☐	☐	☐
14. Synthetic biology primarily involves:		Observing natural ecosystems only	Designing new biological systems	Study of fossils	Classifying organisms	☐	☐	☐	☐
15. In given diagram part iii shows:		A band	H zone	I Band	Sarcomere	☐	☐	☐	☐
16. Plasmids are used as vectors because they:		Only exist in viruses	Cannot carry foreign genes	The destroy DNA of virus	Have an origin of replication	☐	☐	☐	☐
17. For grouped data, Mean=		$\frac{\sum x}{f}$	$\frac{\sum fx}{\sum f}$	$\frac{\sum f}{f}$	$\frac{\bar{x}}{f}$	☐	☐	☐	☐



BIOLOGY HSSC-II

40

Time allowed: 2:35 Hours

Total Marks Sections B and C: 68

SECTION – B (Marks 42)

Q. 2 Answer the following parts briefly.

14x3=42

(i)	Make a comparison between functions of Glomerular capillaries and peritubular capillaries. (Three points)	1x3	OR	How is emphysema caused? What impact does it have on the body?	1.5 + 1.5
(ii)	Identify the ventilation phase. Briefly explain this phase.		03	OR	Identify the parts A, B, and C in the diagram and write one function of each.
(iii)	a. Describe the role of calcium ions in muscle contraction. b. What happens to sarcomere during contraction?	1.5 + 1.5	OR	a. How does synthetic biology differ from traditional genetic engineering? b. Briefly state any one application of synthetic biology.	2+1
(iv)	Calculate Median and Mean of given data: 3, 21, 6, 18, 9, 15, 12.	1+2	OR	Compare arteries and veins. (any three points)	03
(v)	What are monoclonal antibodies? State two advantages of monoclonal antibody over conventional drugs.	03	OR	What is sprain? Enlist its symptoms. How it can be managed?	1x3
(vi)	Briefly describe synapse. Draw its labeled diagram.	1+2	OR	How skin prevents the invasion of pathogens in human body? (any three points)	1x3
(vii)	Briefly describe osteoporosis. Mention its causes and symptoms.	1x3	OR	State the role of baroreceptors and volume receptors in regulating the blood pressure.	03
(viii)	What is nerve impulse? How it is generated in neurons?	03	OR	How are fats absorbed in small intestine?	03
(ix)	Name the hormones secreted by anterior pituitary.	03	OR	Compare ectotherms and endotherms. Give one example of each.	03
(x)	What is complement system? How complement proteins destroy bacteria? (Any two mechanisms briefly)	1x3	OR	Highlight any three harmful effects of ocean acidification.	1x3
(xi)	What is plasmid? How plasmid acts as vectors in genetic engineering?	1+2	OR	Briefly explain the significance of selective re-absorption during urine formation.	03
(xii)	State any three roles of liver.	1x3	OR	Write names and briefly discuss any two species that have gone extinct.	1.5 + 1.5
(xiii)	Differentiate between homologous and analogous structures.	1.5 + 1.5	OR	Associate the effects of drug addiction and tolerance with the functioning of the nervous system.	1.5 + 1.5
(xiv)	Briefly describe two mechanisms used by the human body to lose heat.	1.5 + 1.5	OR	Briefly explain the working of antivirals and antiretrovirals.	1.5 + 1.5

SECTION – C (Marks 26)

Note: Attempt the following questions.

Q.3	a. Discuss any three impacts of climate change on plants. b. How are animals affected by climate change? (Four points)	3+4	OR	How the negative feedback loop help body in regulating homeostasis? Explain with a suitable example.	07
Q.4	Describe the location and function of pancreas as an endocrine gland.	3+3	OR	Elaborate the steps of recombinant biotechnology taking the example of production of insulin.	06
Q.5	a. Draw labelled diagram of human stomach. b. Explain chemical digestion process in human stomach.	3+4	OR	a. Describe the role of B-cells in antibody-mediated immunity. b. Draw the structural model of an antibody molecule.	4+3
Q.6	Describe the ways of CO ₂ transportation in blood: a. In form of HCO ₃ ⁻ b. As carboxyhaemoglobin c. Being dissolved in blood plasma.	3+2 +1	OR	How ammonotelic, ureotelic and uricotelic animals manage the excretion of waste materials in their habitat? Elaborate.	2x3